Knowledge-Grounded Retrieval-Augmented Generation for Regional Situation Awareness

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Human and Biological Systems — Warfighter Decision Superiority and Readiness

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Problem. Regional situation awareness requires turning fast-moving news, field reports, and public statements into a coherent account. Standard retrieval-augmented generation (RAG) can find relevant text but still confuse event timing, merge conflicting claims, or repeat a false report as though it were confirmed. For example, after flooding closes a coastal access route, later updates may contradict early reopening claims; some messages may also be deliberately misleading. Reliable RAG must reason about how events and source claims relate, not just retrieve more passages.

Central hypothesis. Grounding RAG in structured, evolving event knowledge and fine-grained evidence feedback will improve factual support, reasoning consistency, and adaptation to new information. We will investigate adversarial deception as a critical robustness challenge, while making accurate regional knowledge synthesis the primary research objective.

Research plan.

- **Integrate evolving regional knowledge.** Extract entities, events, relations, and timestamps from heterogeneous reports; preserve source provenance and corrections. An updateable event representation will connect related reports and distinguish independent corroboration, unresolved contradiction, and duplicated claims.
- **Generate and verify grounded situation assessments.** Use structured event knowledge to guide retrieval and multi-step synthesis. Break an assessment into checkable assertions about events, locations, and temporal relationships, then use fine-grained feedback to correct unsupported claims. We will study when retrieval and verification should be targeted to a consequential uncertainty.
- **Evaluate adaptation and resistance to deception.** Build controlled report streams with known event histories, delayed corrections, source overlap, and planted misleading content. Study whether the same grounding methods that improve routine summaries also withstand coordinated false reporting. Use scenario variations to support what-if analysis and information-environment exercises.

Evaluation and expected outcomes. GDELT event streams, public disaster-response timelines, and synthetic time-stamped report sets will test updating and robustness. We will measure claim support, event consistency, timeliness, sensitivity to deception, and cost. Baselines include standard RAG and iterative retrieval with LLM self-critique.

Naval relevance. For a Navy or Marine Corps team planning humanitarian assistance in a littoral region, conflicting updates about damaged roads and port access can change the local situation picture. Knowledge-grounded RAG could produce traceable updates while identifying what has changed, what is disputed, and where deception may be present. The work directly matches regional situation awareness and AI-enhanced RAG, with relevance to operations in the information environment.

Prior work and PI qualifications. Our QA-based event extraction (Du and Cardie, 2020), FaithScore (Jing et al., 2024), and FGAIF (Jing and Du, 2025) provide foundations for event representation and fine-grained grounding feedback. The PI holds a 2024 NSF CAREER award and led integration of the RESIN team's DARPA KAIROS delivery system as a postdoc; the team achieved top performance among participants.